

Sender MAC Address - Belonging to system which replies to the ARP request Updated by the system - 00:21:97:88:28:21
 Sender IP Address - Belonging to system which replies to the ARP request - 192.168.51.1
 Target MAC Address - Source MAC of ARP request packet - 7c:05:07:ad:42:53
 Target IP Address - Source IP address of the ARP request packet - 192.168.51.208

Wireshark displays the ARP reply under the 'Info' box as: 192.168.51.1 is at 00:21:97:88:28:21.

Thus, with the help of an ARP request and reply, system 192.168.51.208 has detected the MAC address belonging to 192.168.51.1.

Dynamic Host Configuration Protocol (DHCP)

This protocol saves a lot of time for network engineers by offering a unique dynamic IP address to a system without an IP address, which is connected in a network. This also helps to avoid IP conflicts (the use of one IP address by multiple systems) to a certain extent. The computer users also benefit by the ability to connect to various networks without knowing the corresponding IP address range and the unused IP address.

This DHCP protocol consists of four phases—DHCP discover, DHCP offer, DHCP request and DHCP ACK. Let us understand the protocol and interpret how these packets are seen in Wireshark.

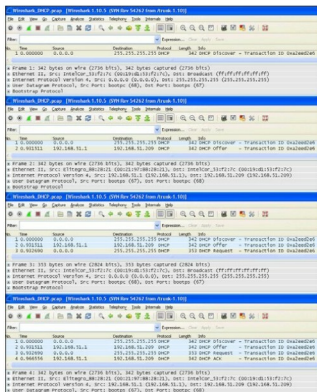


Figure 4: DHCP protocol

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Announcing the Launch of the 'Ask the Expert' section



Over the past two decades, the Linux and open source movement has been getting more prominence. Open Source For You (formerly known as Linux For You) was one of the first magazines to recognise the trend and launch a platform for professionals to get and share information about open source and Linux.

With the open source movement surging forward across the globe, its deployment in various verticals has increased, thus generating new and more complex challenges for systems administrators and IT managers.

To address these challenges, we at OSFY have launched the 'Ask the Expert' section, dedicated to answering queries from IT managers and sysadmins on the following topics

1. Network infrastructure
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3. Messaging (email, audio/video-chat, instant messaging)
4. Network architecture
5. Network storage
6. Business applications



E-mail your queries to askexpert@efy.in, and get the answers from our team of experts.

From the OSFY team

